Kalhara Tennakoon

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Professional Summary

First Class Computer Science (Software Engineering) graduate and Software Engineering batch top with industry experience at IFS R&D International in Kubernetes-native CI/CD automation, container security, and cloud-native engineering. Experienced in building full-stack applications, REST APIs, ML forecasting systems, and AI applications using local LLMs, RAG, and vector search, alongside GitOps-based Kubernetes deployments of ML applications. Proficient in Python, JavaScript, Docker, Kubernetes, CI/CD, and AI-assisted engineering using modern coding assistants. Interested in building reliable, scalable, and intelligent software systems across Software Engineering, Cloud, DevOps, AI/ML, and MLOps.

Technical Skills

Programming: Python, Java, JavaScript, Go, C, Bash/Shell Scripting
Software Engineering: React, Node.js, Express, Flask, FastAPI, REST APIs, Full-Stack Development, API Development, JWT Authentication
AI / Machine Learning: Scikit-learn, Pandas, Prophet, pgvector, Ollama, RAG, Local LLMs, Data Cleaning, Exploratory Data Analysis, Feature Engineering, Time-Series Forecasting
AI-Assisted Engineering: ChatGPT, Claude, Google Gemini, GitHub Copilot, Cursor; AI-assisted coding, debugging, refactoring, test generation, code review, Python/ML development, RAG/LLM application development, and DevOps troubleshooting
Cloud & DevOps: Kubernetes, Docker, Tekton, Argo CD, Kustomize, GitHub Actions, Azure DevOps, Trivy, CI/CD, Infrastructure as Code, GitOps, Container Security
Cloud Platforms: Microsoft Azure, AWS, Google Cloud Platform, AKS, GKE, Rancher
Databases: PostgreSQL, MySQL, MongoDB
Tools & Practices: Git, GitHub, Jira, Agile, Scrum, Technical Documentation, Automated Testing

Professional Experience

IFS R&D International (Pvt) Ltd **Colombo, Sri Lanka**
DevOps Engineering Intern *Jul 2024 – Oct 2024*

- Automated Kubernetes-native CI/CD workflows using Tekton, streamlining deployment processes for critical services.
- Integrated Behave automated testing into Tekton CI/CD pipelines, reducing manual debugging effort by approximately 30% and improving deployment reliability.
- Optimized pipeline configurations and resolved recurring build and deployment issues to improve deployment stability and engineering efficiency.
- Conducted knowledge-sharing sessions on testing best practices to support team collaboration and code quality.

Software Engineering Intern *Jul 2020 – Dec 2020*

- Developed a Docker and Kubernetes-based security scanning solution for container platforms and vulnerability assessment workflows.
- Integrated security scanning tools into CI/CD pipelines to improve vulnerability detection across deployment stages.
- Diagnosed deployment and runtime issues and authored technical documentation supporting system reliability and onboarding.
- Worked with cloud-native technologies and containerized environments to strengthen secure software delivery practices.

Projects

VetCare One — Smart Veterinary Clinic Management System
React, Node.js, Express, PostgreSQL, Python, Flask, Scikit-learn, pgvector, Ollama, SwiftUI Oct 2025 – Aug 2026

- Built a full-stack veterinary clinic management platform covering appointments, medical records, billing, inventory, role-based access control, and JWT authentication.
- Developed a locally hosted RAG assistant using Ollama and pgvector, combining semantic retrieval with exact SQL queries for reliable access to structured clinic data.
- Implemented human-in-the-loop controls requiring confirmation for database write operations.
- Developed ML models for disease prediction, sales forecasting, and inventory demand forecasting.
- Built a pet-owner portal and SwiftUI companion iOS application to extend access beyond the clinic platform.
- Used AI-assisted coding and debugging workflows to support Python development, RAG integration, application testing, and refinement of AI-powered features.
- Selected among the Top 10 finalists of the Ascentic AI Launch Pad 2026.

SolarCast — Solar Energy Forecasting Application with GitOps Deployment
Python, FastAPI, React, Prophet, Docker, Kubernetes, Kustomize, Argo CD, GitHub Actions Apr 2026 – Oct 2026

- Built a full-stack machine learning application with a FastAPI backend and React dashboard that forecasts monthly solar generation, grid export, and expected payouts from residential inverter data.
- Developed a time-series forecasting pipeline using Meta Prophet with 90% confidence ranges, tested on unseen months with a chronological 70/30 split: about 8% error (MAPE) against 9.5% for a same-month-last-year baseline.
- Built a GitHub Actions CI/CD pipeline with linting, 39 automated tests, Trivy vulnerability scans, and end-to-end smoke tests, publishing multi-stage, non-root Docker images to Docker Hub.

- Deployed to Kubernetes dev and prod environments using Kustomize and Argo CD: merges to main roll out to dev, prod is promoted by pull request, and rollbacks are Git reverts.
- Applied AI-assisted development to support Python implementation, debugging, test-case development, and review of containerized application and CI/CD configurations.

Predicting Course Difficulty from Student Evaluation Responses
Python, Scikit-learn, Pandas, Matplotlib, Seaborn, SPSS Feb 2025 – Jul 2025

- Developed supervised machine learning models to predict perceived course difficulty using student evaluation datasets.
- Performed data preprocessing, exploratory data analysis, feature engineering, and model evaluation using cross-validation.
- Applied statistical significance testing and model interpretation to identify instructor-related attributes associated with perceived course difficulty.

Kubernetes Cluster & Container Image Security Scanner
Kubernetes, Docker, Go, Argo CD, Rancher, Azure DevOps Jul 2020 – Dec 2020

- Developed a containerized security scanning solution for Kubernetes environments to detect vulnerabilities in container images and cluster configurations.
- Integrated automated security scanning workflows into CI/CD pipelines to strengthen secure deployment practices and vulnerability detection coverage.
- Utilized cloud-native platforms and GitOps tooling to support scalable, secure, and reliable Kubernetes operations.

EDUCATION

Kingston University, London, United Kingdom **2026**
BSc (Hons) Computer Science (Software Engineering) — First Class Honours
Studied in Sri Lanka through ESU.

- Batch Top — Software Engineering, ESU Kandy

CERTIFICATIONS

Cloud, DevOps & Platform Engineering

- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: Azure Data Fundamentals (DP-900)
- Introduction to Kubernetes — The Linux Foundation
- Containers & Kubernetes Essentials — IBM
- GitHub Foundations — GitHub
- Introduction to GitOps

Artificial Intelligence & Data

- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Google AI Professional Certificate

HONOURS & AWARDS

Top 10 Finalist – Ascentic AI Launch Pad 2026
Selected among 700+ applicants and advanced to the final round.

Batch Top – Software Engineering, ESU Kandy 2026
Ranked first in the Software Engineering cohort.

Two-Time Dan Kohn Scholarship Recipient – KubeCon + CloudNativeCon Europe 2022 & 2025
Awarded scholarships recognizing contributions to the cloud-native ecosystem.

First Runner-Up – Hack:Bit Hackathon 2019
Secured 2nd place in a competition organized in collaboration with Microsoft and Sarvodaya Fusion.

Top 15 Finalist – HaXmas Hackathon 2018
Selected among 75+ competing teams.

Cisco NetRiders / Packet Riders
Ranked 4th in Sri Lanka and 118th in the Asia Pacific & Japan region.

TECHNICAL WRITING

Technical Writer — Medium Dec 2021 – Present

- Authored 50+ technical articles covering DevOps, cloud-native technologies, Kubernetes, Tekton, Docker, Go, CI/CD, and software engineering.
- Generated more than 150,000 views across published technical content.
- Published technical content through platforms including Level Up Coding and FAUN.dev.

COMMUNITY ENGAGEMENT

Volunteer — STEMUp Educational Foundation Aug 2018 – Dec 2020

- Contributed to initiatives promoting STEM education and digital literacy among students and youth communities.